ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Density-based Clustering

Session 25 and 26

BSDSBA 2028

22 April 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Due this Week – R12 (Wed April 22), ICA11 (Wed April 22)

Upcoming – E4S2 (Thu April 30), E8 (Thu April 30), A2 – Data Mining 1 (Mon May 4), A3 – Data Mining 2 (Fri May 8), MP2 (TBA)

Final Project – Slides Submission (Tues May 5); Public Presentation (Thu May 7)

Final Exam – Wednesday, May 6, 2026; Coverage: Session 15 onwards (from IR to Clustering)

Session 25 and 26 – Density-Based Clustering

Gameplan

Session 25

Density-Based Clustering: DBSCAN

Clustering Evaluation

Session 26

Applied Example of DBSCAN & Clustering Evaluation

In-Class Activity 12 – Density-based Clustering

Session 27 and 28 – Lab Session & Course Wrap-Up

Gameplan

Session 27

Per LT Consultation (on Final Project or MP2)

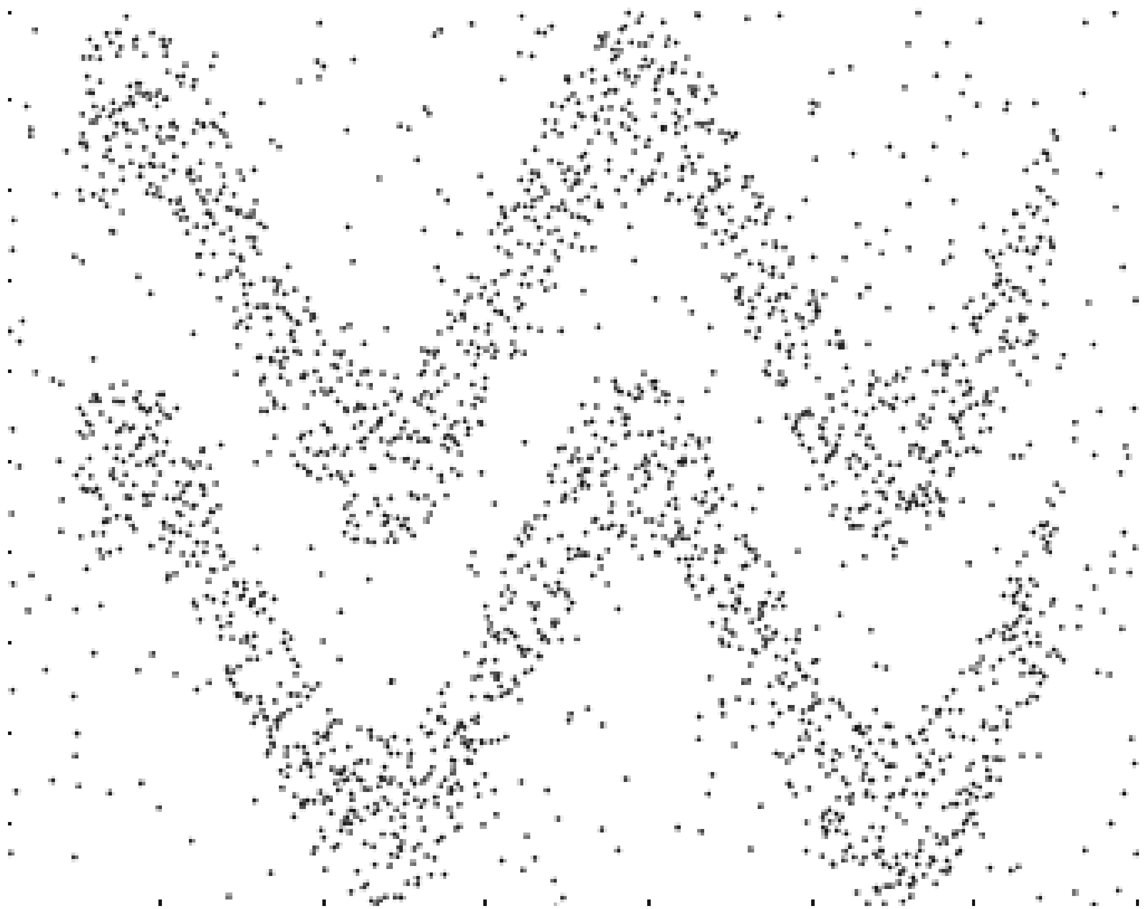
Schedule for each team will be determined later

Session 28

Course Wrap-Up and Way Forward

Density-based Clustering Methods

Density-based Clustering Methods

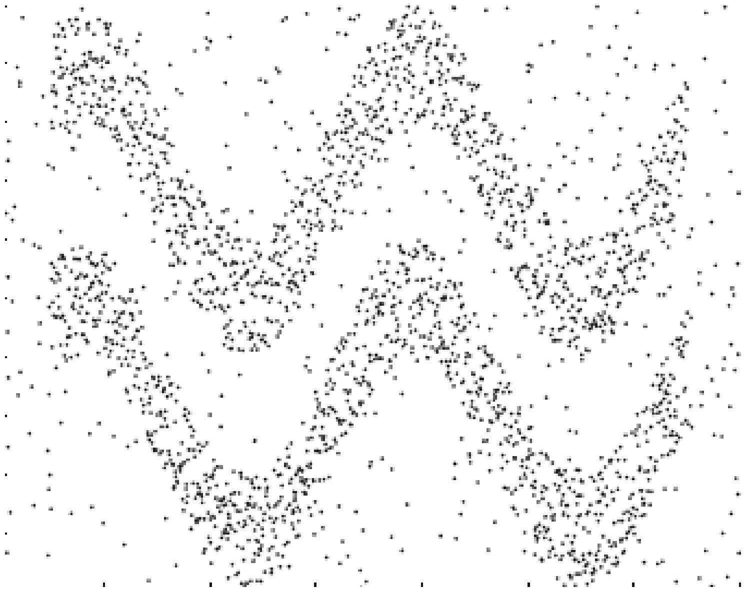


*Aggarwal, C. C. (2015). **Data mining: the textbook**. Chapter 6.*

Density-based Clustering Methods

What is Density-based Clustering?

A density-based clustering method identifies clusters as dense connected regions of data points separated by sparser areas.



Aggarwal, C. C. (2015). ***Data mining: the textbook****. Chapter 6.*

Density-based Clustering Methods

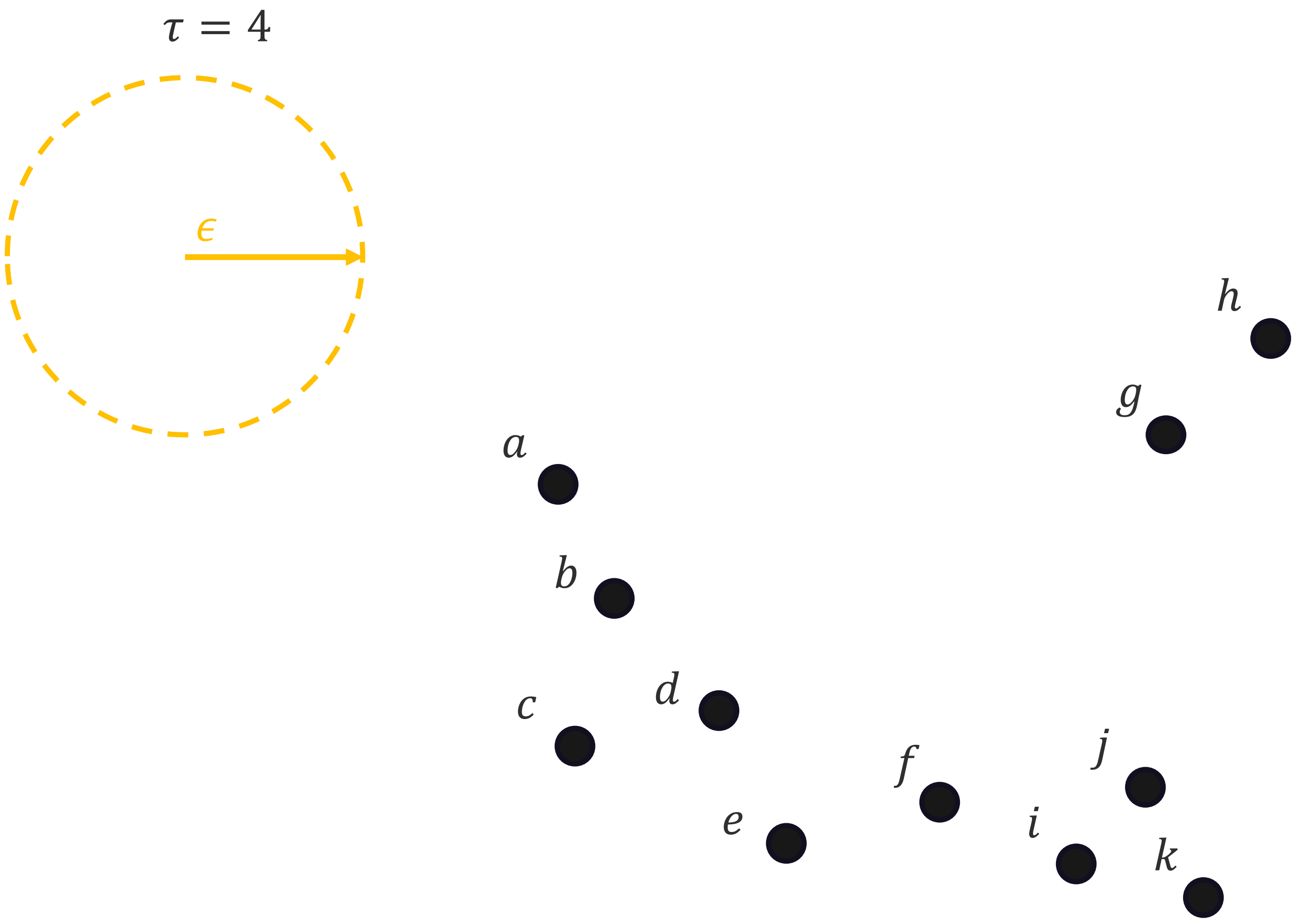
What is DBSCAN?

DBSCAN, shorthand for **Density-Based Spatial Clustering of Applications with Noise**, finds *core objects* then connects them and their *neighborhoods* to form dense clusters.

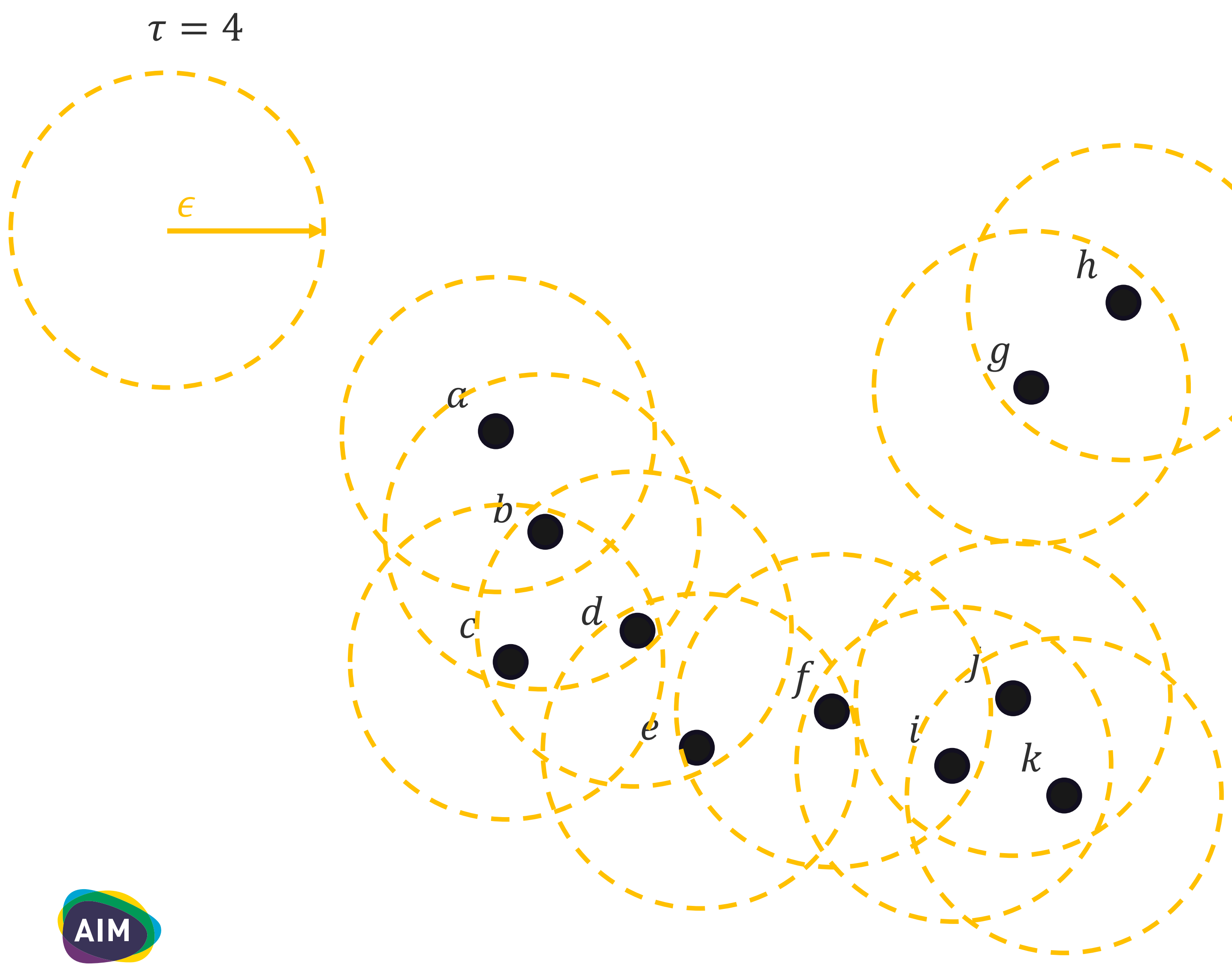
DBSCAN requires two parameters – ϵ to specify the radius of the neighborhood, and minimum number of points τ which specifies the density threshold of dense regions.

Aggarwal, C. C. (2015). ***Data mining: the textbook****. Chapter 6.*

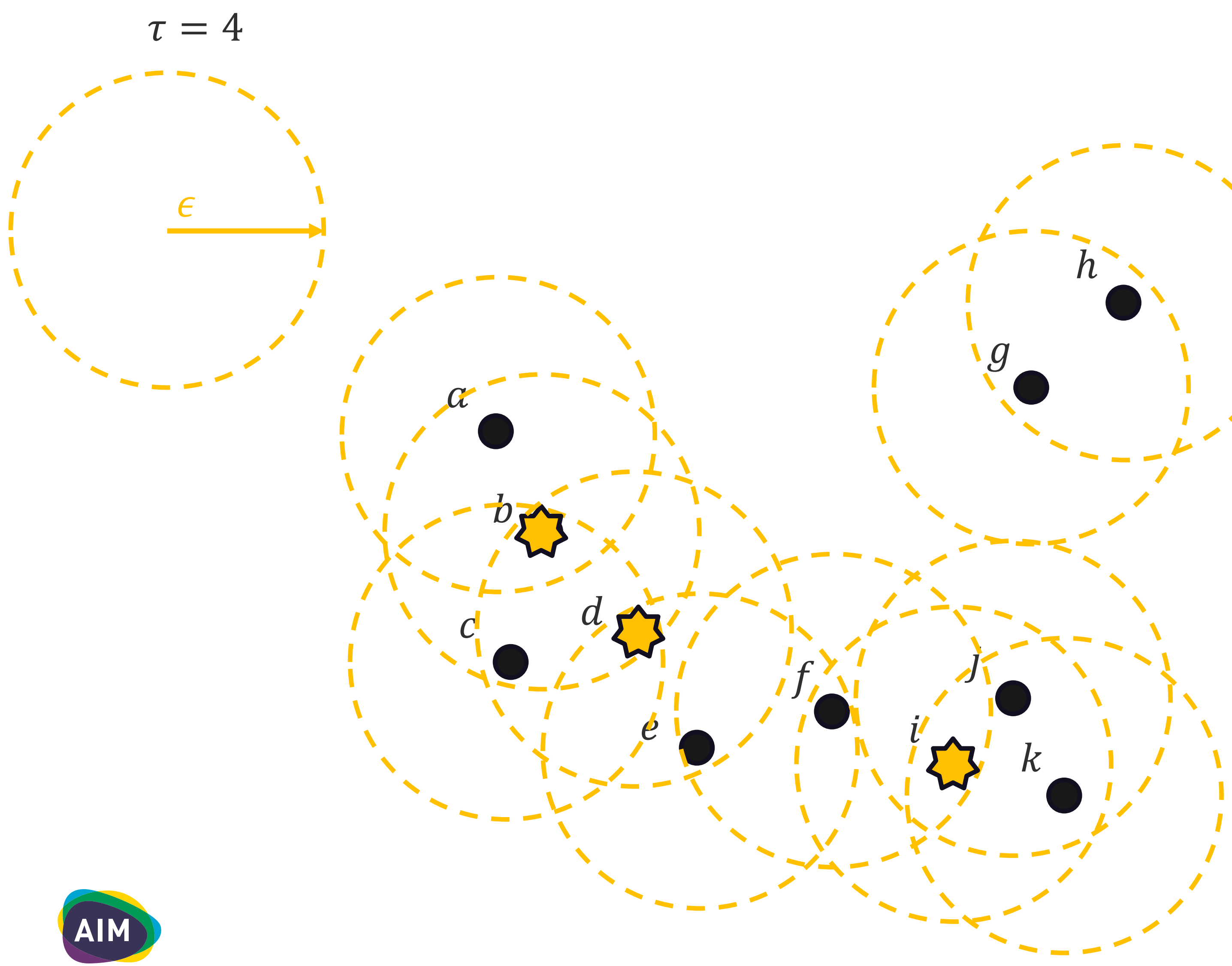
Density-based Clustering Methods



Density-based Clustering Methods

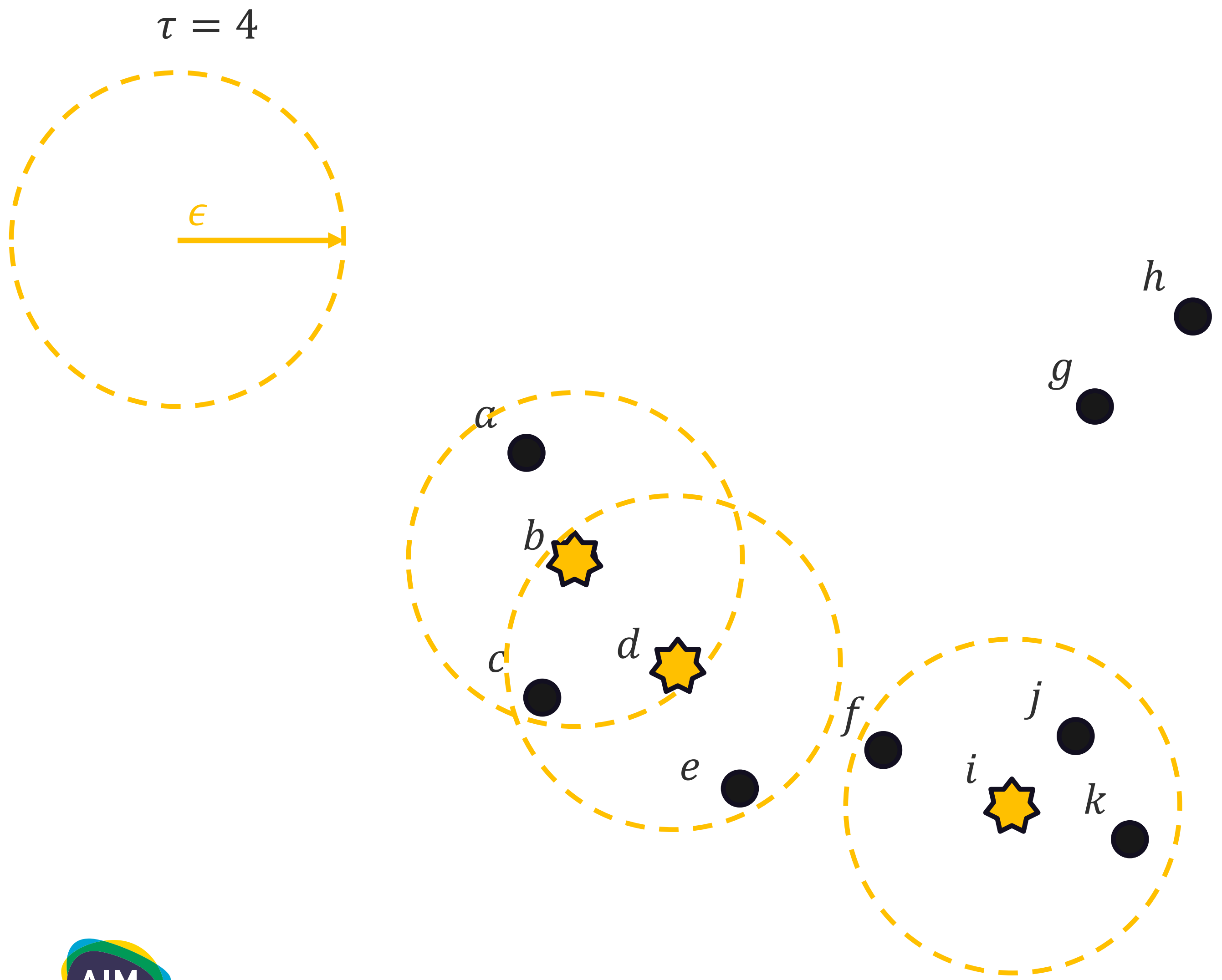


Density-based Clustering Methods



A **core point** contains at least τ points within ϵ

Density-based Clustering Methods



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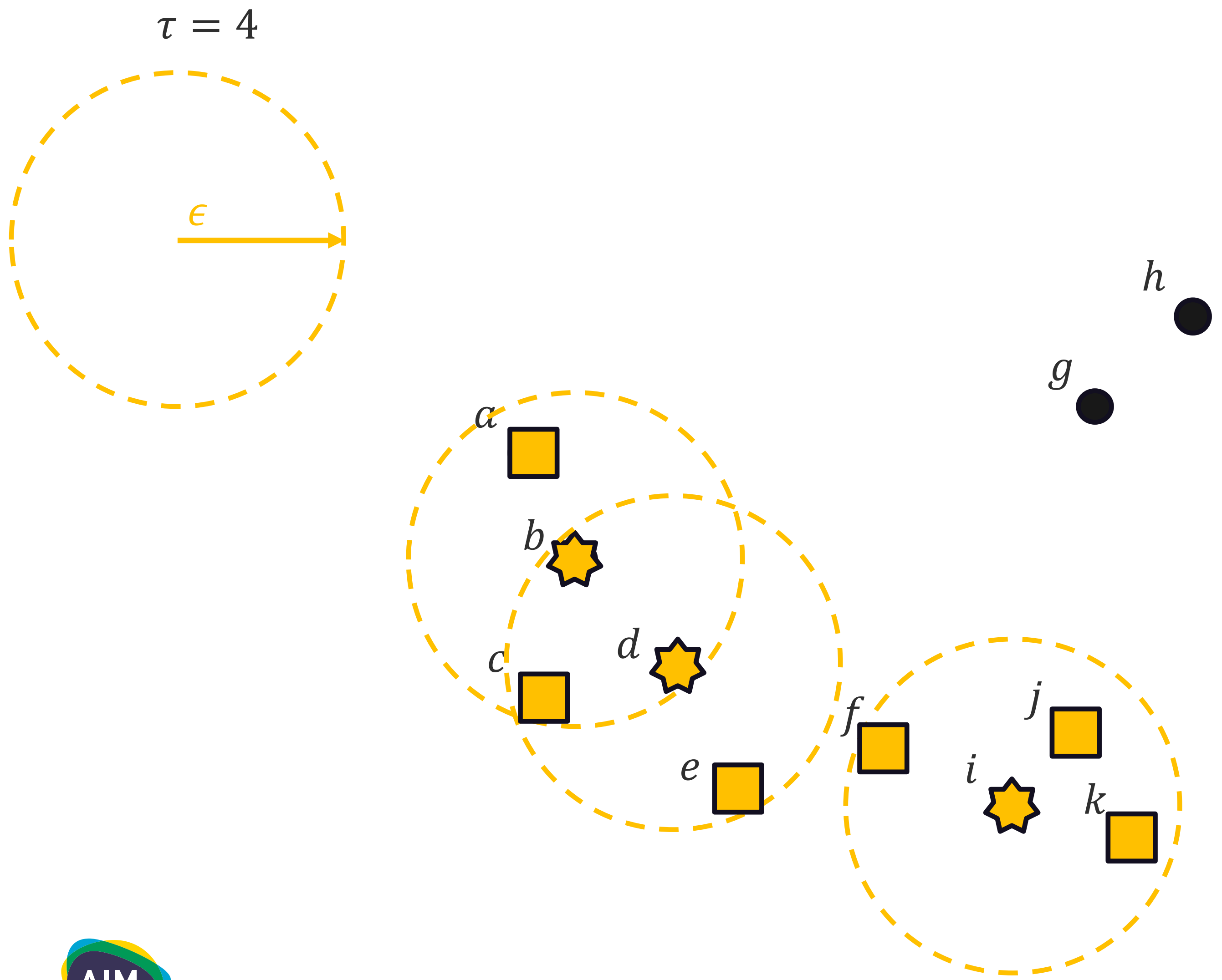
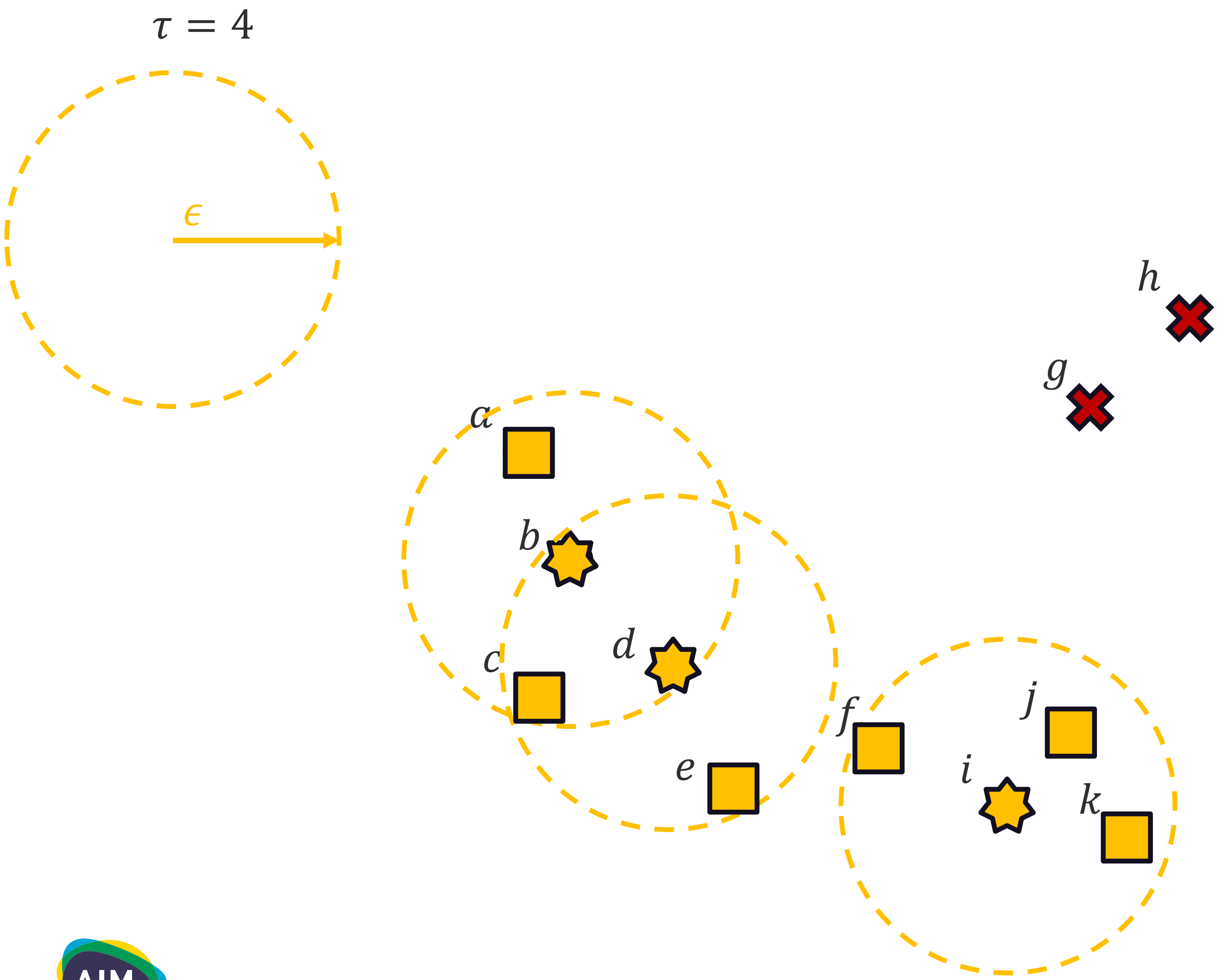
Density-based Clustering Methods



A **core point** contains at least τ points within ϵ

A **border point** contains less than τ points within ϵ but contains a core point within ϵ

Density-based Clustering Methods

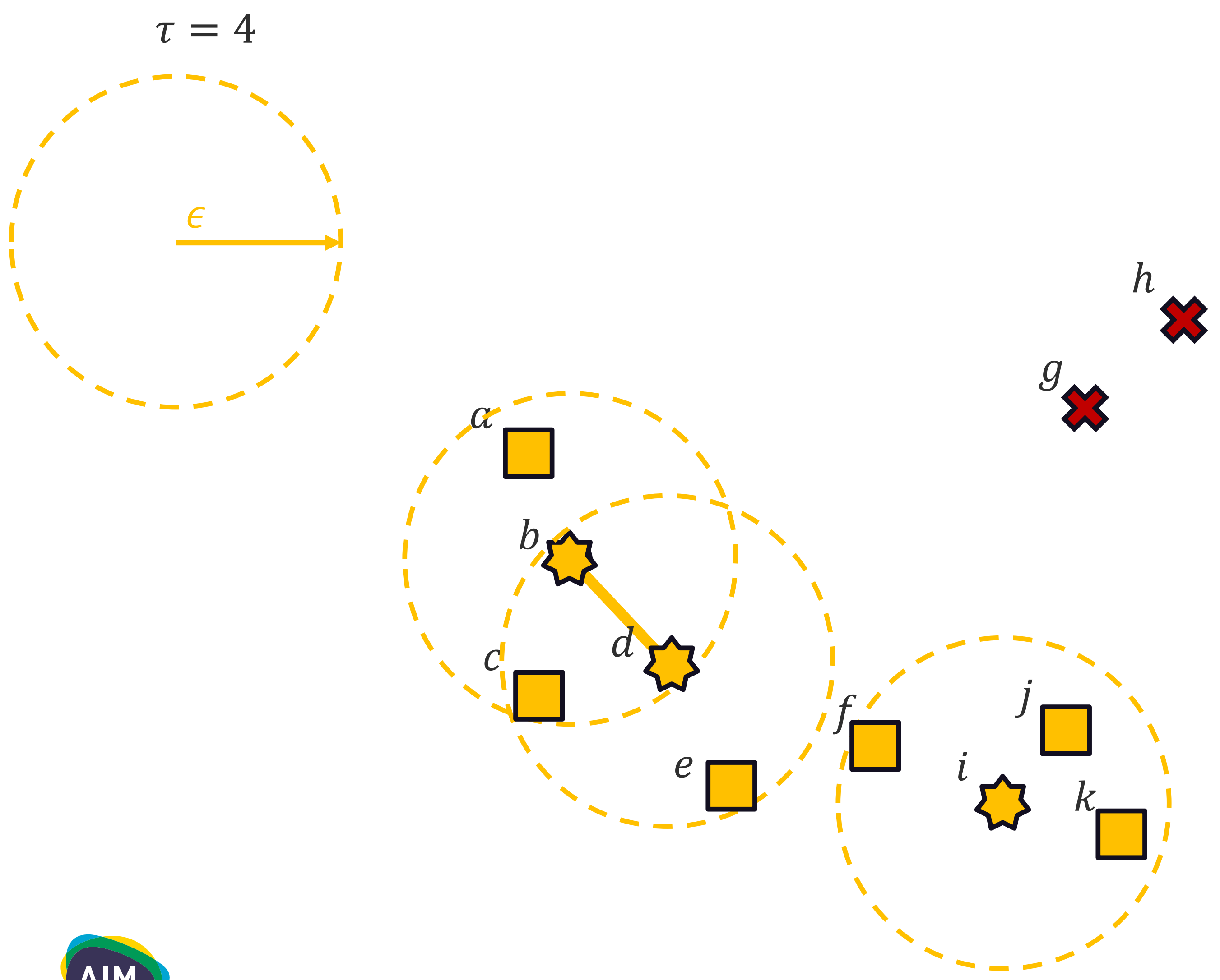


A **core point** contains at least τ points within ϵ

A **border point** contains less than τ points within ϵ but contains a core point within ϵ

A **noise point** is neither a core nor a border point

Density-based Clustering Methods



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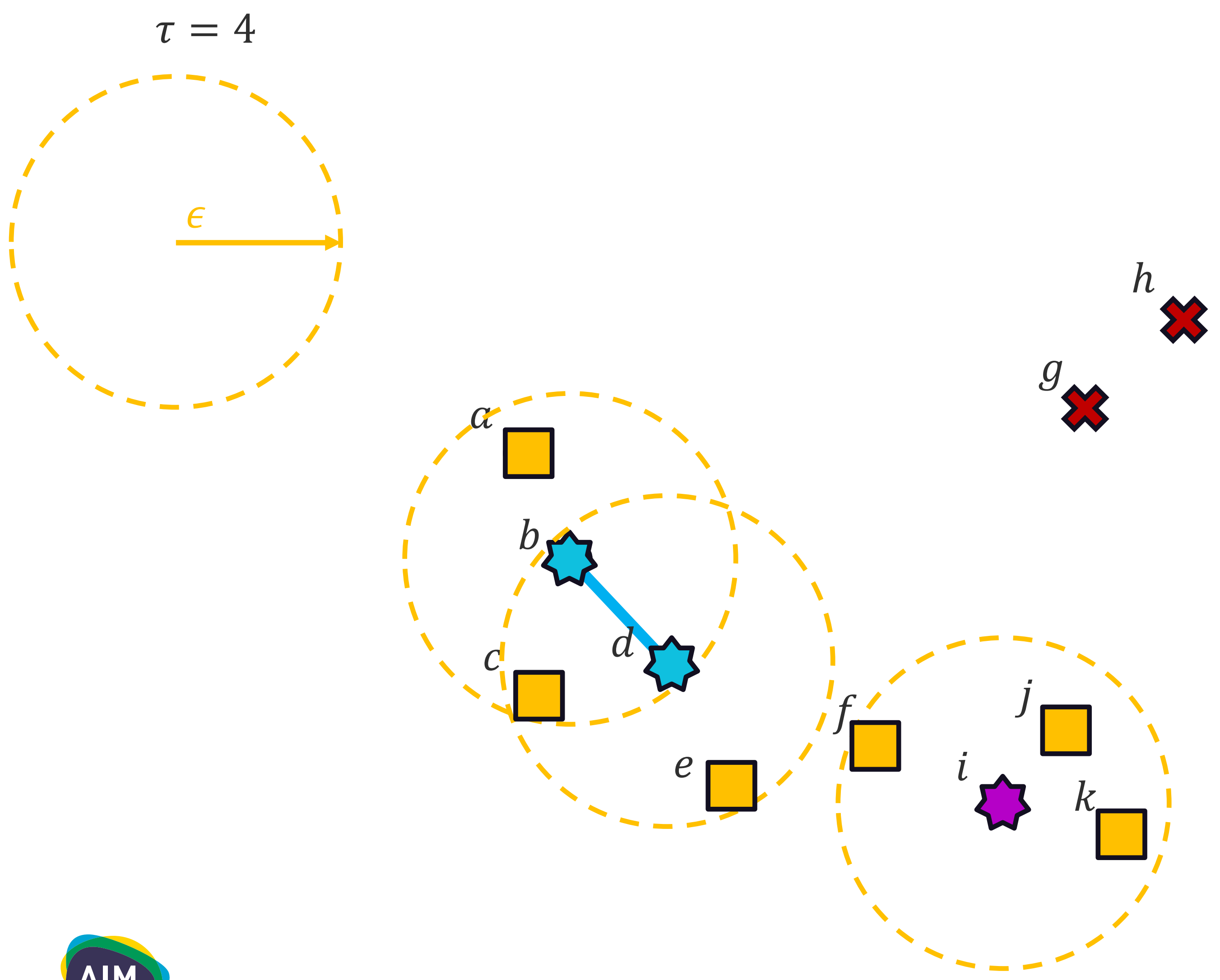
Density-based Clustering Methods



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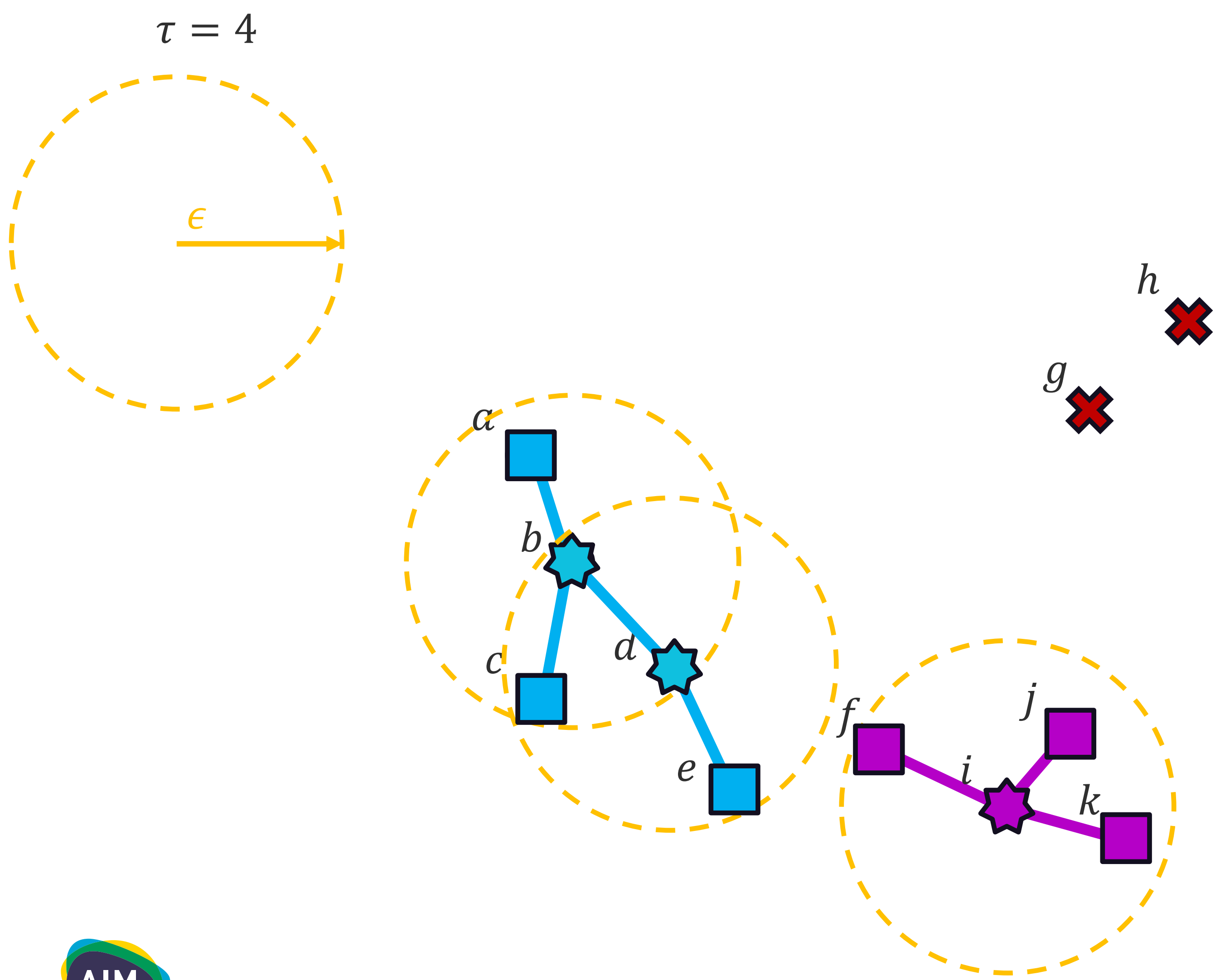
Density-based Clustering Methods



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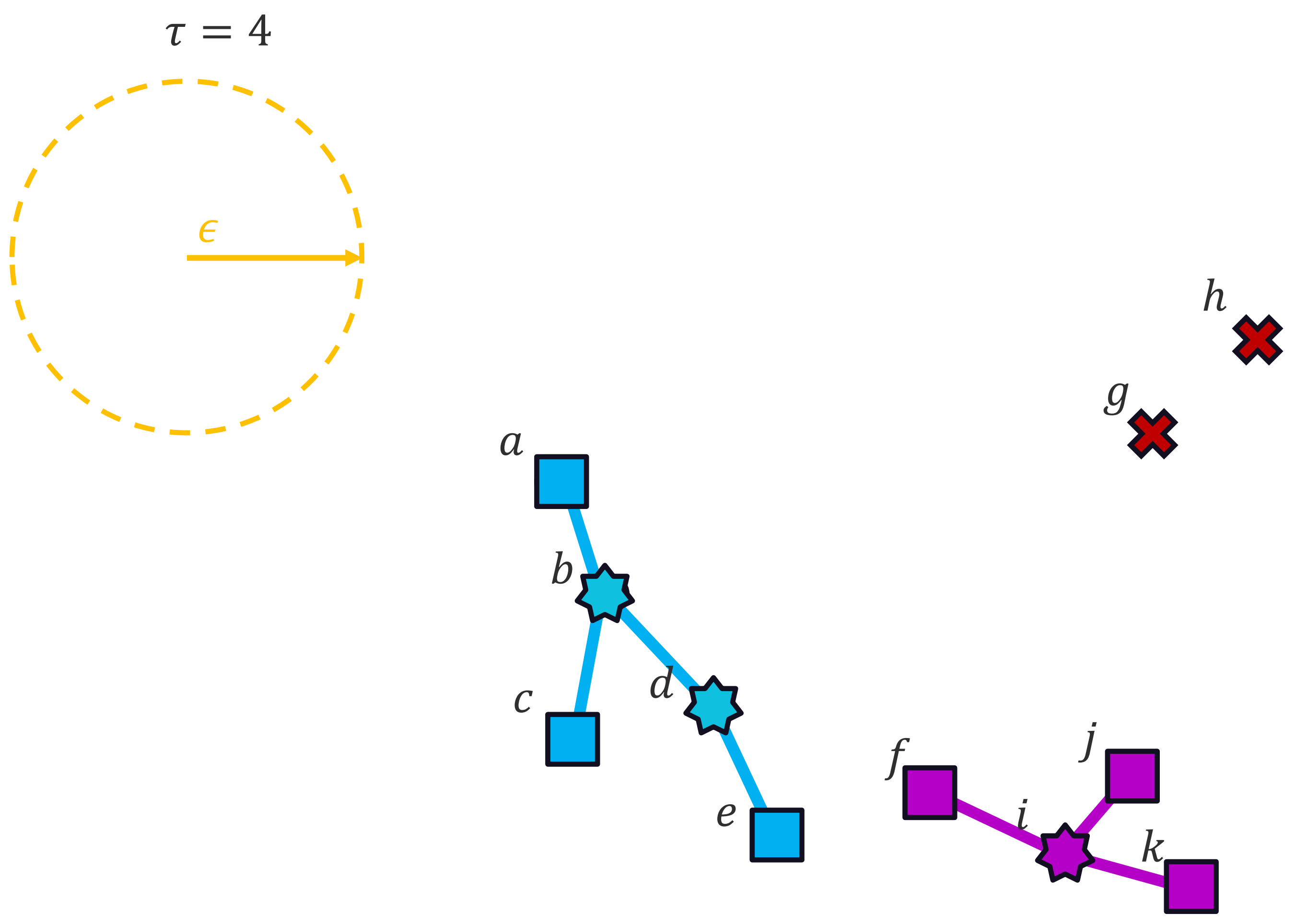
Density-based Clustering Methods



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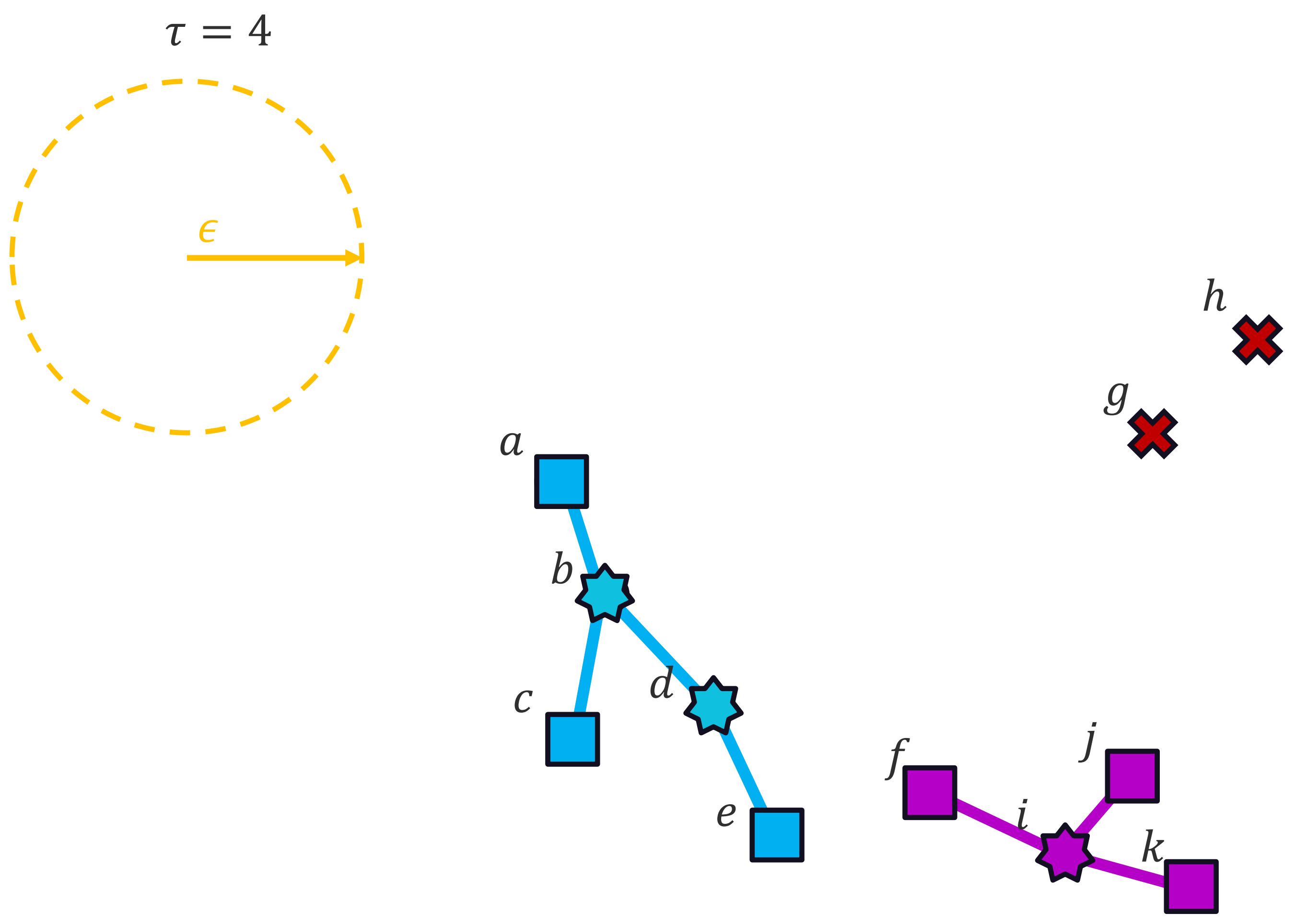
Density-based Clustering Methods



A **core point** contains at least τ points within ϵ

A **border point** contains less than τ points within ϵ but contains a core point within ϵ

A **noise point** is neither a core nor a border point

Clusters: $\{a, b, c, d, e\}$, $\{f, i, j, k\}$

Noise: $\{h, g\}$

Density-based Clustering Methods

Algorithm for DBSCAN

Given a data set $\mathcal{D}$, radius ϵ, density τ

start

Determine core, border, and noise points of $\mathcal{D}$ at level (ϵ, τ)

Create a graph in which core points within ϵ are connected;

Determine connected components in graph;

Assign each border point to connected component with which it is best connected;

return points in each connected component as a cluster;

end

Density-based Clustering Methods

Advantages

- DBSCAN method can discover clusters of arbitrary shape.
- Highly effective when clusters are separated by low-density regions.
- DBSCAN does not require a number of clusters as an input parameter.
- DBSCAN also identifies the noise and outlier data points.

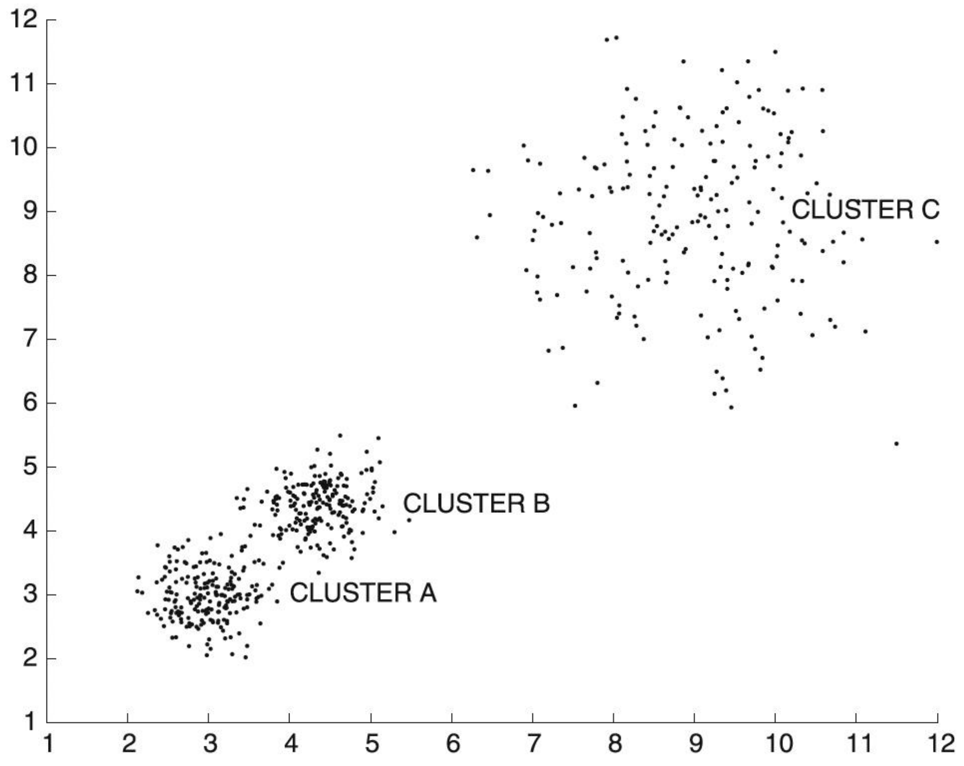
Density-based Clustering Methods

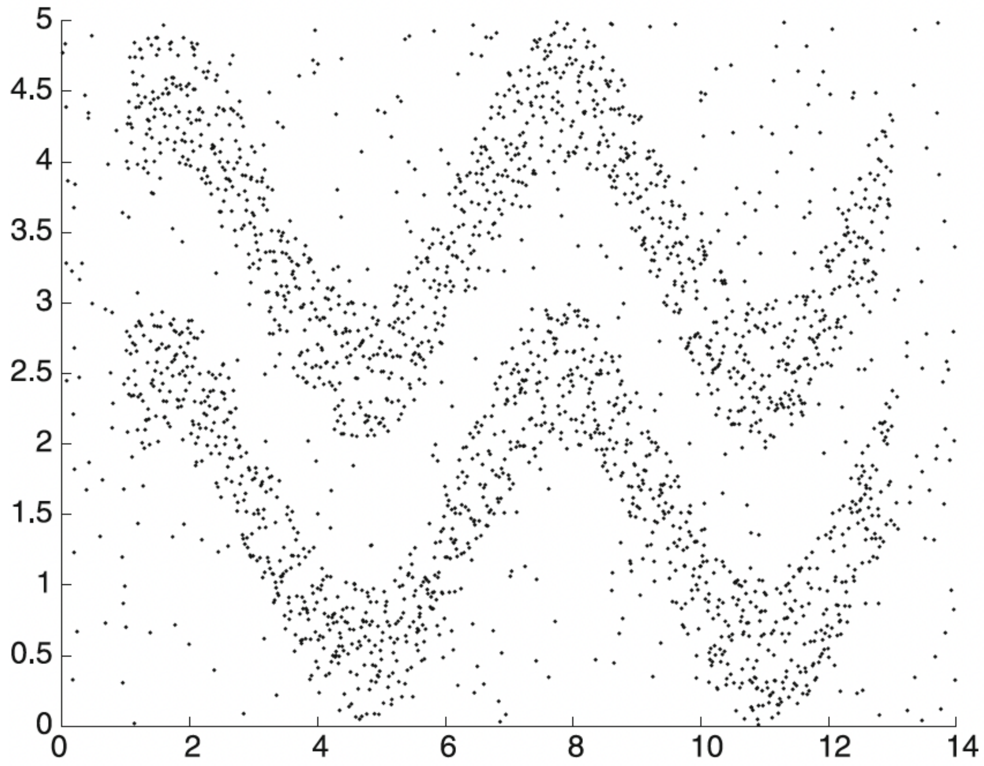
Disadvantages

- Result is sensitive to the choice of ϵ and τ
- Struggles with varying density clusters
- Performance can degrade in high-dimensional spaces
- Not ideal for datasets with significant overlap in density

Density-based Clustering Methods

Disadvantages





Clustering Evaluation

Clustering Evaluation

What are ways of measuring the quality of a clustering?

Extrinsic methods – comparing the clustering result against a ground truth and measure.

Intrinsic methods – evaluates the goodness of a clustering without any ground truth.

Han, J., Pei, J., & Tong, H. (2023). ***Data mining: concepts and techniques****. Morgan Kaufmann.*

Clustering Evaluation

What makes a good clustering? (Intrinsic Methods)

Fundamentally, a good clustering procedure should produce **compact** and **well-separated** clusters. Points in the same cluster should be close together, meanwhile, points not belonging in the cluster should be far from points in the cluster.

Clustering Evaluation

Silhouette Coefficient

For a dataset, D, of n objects, D is partitioned into k clusters $C_1, \ldots, C_k$. For each object $o \in D$. Calculate $a(o)$ as the average distance between o and all other objects in the cluster of o. Similarly, $b(o)$ is the minimum average distance from o to all clusters to which o does not belong.

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Clustering Evaluation

Silhouette Coefficient

For $o \in C_i$ $(1 \leq i \leq k)$

$$a(o) = \frac{\sum_{o' \in C_i, o \neq o'} dist(o, o')}{|C_i| - 1}$$

$$b(o) = \min_{C_j: 1 \leq j \leq k, j \neq i} \left\{ \frac{\sum_{o' \in C_j} dist(o, o')}{|C_j|} \right\}$$

AIM

Han, J., Pei, J., & Tong, H. (2023). **Data mining: concepts and techniques**. Morgan Kaufmann.

Clustering Evaluation

Silhouette Coefficient

The **silhouette coefficient** of $\boldsymbol{o}$ is defined as

$$s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}}$$

Han, J., Pei, J., & Tong, H. (2023). ***Data mining: concepts and techniques***. *Morgan Kaufmann.*

Clustering Evaluation

Davies-Bouldin Index

Compares the mean similarity of a cluster to its most similar cluster. Define s_i as the average distance between each point $o \in C_i$ and the centroid $\overline{o_i}$ of C_i. Also define, d_{ij} to be the distance between cluster centroids $\overline{o_i}$ and $\overline{o_j}$. The similarity between two clusters is defined as:

$$R_{ij} = \frac{s_i + s_j}{d_{ij}}$$

Clustering Evaluation

Davies-Bouldin Index

The **Davies-Bouldin Index** is defined as:

$$DB = \frac{1}{k}\sum_{i} \max_{i \neq j} R_{ij}$$

Clustering Evaluation

Utilizing Ground Truth (Extrinsic Methods)

Matching-based – examines how well clustering results match the ground truth in partitioning the data set.

Information theory-based – compares the distribution between clustering results and the ground truth.

Pairwise-comparison-based – checks the pairwise consistency of the objects in the clustering result.

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Clustering Evaluation

Matching-based Method

Purity

Measures the extent at which clusters contain data points from a single ground-truth class.

Let $D = \{o_1, \dots, o_n\}$ be set of data points, which a clustering method partitions them into $\mathcal{C} = \{C_1, \dots, C_m\}$ clusters. Meanwhile, a ground truth class partitions them into $\mathcal{G} = \{G_1, \dots, G_l\}$ groups.

$$purity = \frac{1}{n}\sum_{i=1}^{m} \max_{j=1}^{l}\{|C_i \cap G_j|\}$$

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Clustering Evaluation

Example: Purity

Given the result of clustering methods $\mathcal{C}_1$ and $\mathcal{C}_2$ and the ground truth from the table below, calculate for the purity of the clustering $\mathcal{C}_1$and $\mathcal{C}_2$.

Object	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*
Ground truth $\mathcal{G}$	G_1	G_1	G_1	G_1	G_1	G_1	G_2	G_2	G_2	G_2	G_3
Clustering $\mathcal{C}_1$	C_1	C_1	C_1	C_1	C_2	C_2	C_3	C_3	C_3	C_3	C_4
Clustering $\mathcal{C}_2$	C_1	C_1	C_2	C_2	C_2	C_3	C_1	C_2	C_2	C_1	C_3

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Clustering Evaluation

Information theory-based Method

Entropy H (as clustering metric)

Treats the groupings of a clustering and ground truth as “states” or possible outcomes for each data point, then compares the average uncertainty level or information associated with the states as expressed by the clustering and the ground truth labels.

$$H(X) = -\sum_{x \in X} p(x) \log p(x)$$

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Clustering Evaluation

Information theory-based Method

Entropy H (as clustering metric)

Conditional entropy of $\mathcal{G}$ given cluster $\mathcal{C}_i$

$$H(\mathcal{G}|\mathcal{C}_i) = -\sum_{j=1}^{l} \frac{|C_i \cap G_j|}{|C_i|} \log \frac{|C_i \cap G_j|}{|C_i|}$$

Conditional entropy of $\mathcal{G}$ given clustering $\mathcal{C}$

$$H(\mathcal{G}|\mathcal{C}) = \langle H(\mathcal{G}|\mathcal{C}_i) \rangle_i = \sum_{i=1}^{m} \frac{|C_i|}{n} H(\mathcal{G}|\mathcal{C}_i)$$

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Clustering Evaluation

Example: Entropy

Given the result of clustering methods $\mathcal{C}_1$ and $\mathcal{C}_2$ and the ground truth from the table below, calculate for the entropy of the clustering $\mathcal{C}_1$ and $\mathcal{C}_2$.

Object	*a*	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*
Ground truth $\mathcal{G}$	G_1	G_1	G_1	G_1	G_1	G_1	G_2	G_2	G_2	G_2	G_3
Clustering $\mathcal{C}_1$	C_1	C_1	C_1	C_1	C_2	C_2	C_3	C_3	C_3	C_3	C_4
Clustering $\mathcal{C}_2$	C_1	C_1	C_2	C_2	C_2	C_3	C_1	C_2	C_2	C_1	C_3

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Clustering Evaluation

Object	a	b	c	d	e	f	g	h	i	j	k
Ground truth $\mathcal{G}$	G_1	G_1	G_1	G_1	G_1	G_1	G_2	G_2	G_2	G_2	G_3
Clustering $\mathcal{C}_1$	C_1	C_1	C_1	C_1	C_2	C_2	C_3	C_3	C_3	C_3	C_4
Clustering $\mathcal{C}_2$	C_1	C_1	C_2	C_2	C_2	C_3	C_1	C_2	C_2	C_1	C_3

$$H(\mathcal{G}|C_i) = -\sum_{j=1}^{l} \frac{|C_i \cap G_j|}{|C_i|} \log \frac{|C_i \cap G_j|}{|C_i|}$$

$$H(\mathcal{G}|\mathcal{C}) = \langle H(\mathcal{G}|\mathcal{C}_i) \rangle_i = \sum_{i=1}^{m} \frac{|C_i|}{n} H(\mathcal{G}|C_i)$$

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Clustering Evaluation

Pairwise comparison-based Method

For each pair of objects $o_i, o_j \in D$ $(1 \leq i, j \leq n, i \neq j)$, if they are assigned to the same cluster/group, the assignment is regarded as positive, and otherwise, negative. Then, depending on the clustering assignment $C(o_i)$ and $C(o_j)$ and ground truth assignments $G(o_i)$ and $G(o_j)$, we have four possible cases.

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

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Clustering Evaluation

Pairwise comparison-based Method: Jaccard coefficient

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

$$J = \frac{TP}{TP + FN + FP}$$

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